

1. Administrative & Study Identification

Full Study Title: A Phase 2, Open-label, Randomized Controlled Trial of BMS-986218 or BMS-986218 Plus Nivolumab in Combination With Docetaxel in Participants With Metastatic Castration-resistant Prostate Cancer **Short Title/Acronym:** CA022-009 Study **Protocol Number:** CA022-009 **Posting ID:** 987628428492287200 **NCT Number:** NCT05169684 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** BMS-986218 (Anti-CTLA4-NF mAb) and Nivolumab (PD-1 Inhibitor) in combination with Docetaxel (ATC Code: L01CD02; Trade names: Docefrez, Taxotere) **Phase of Development:** Phase II **Trial Status:** TERMINATED (Study was terminated early) **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, efficacy, tolerability, and toxicity of docetaxel alone, in combination with BMS-986218, or in combination with nivolumab plus BMS-986218 in men who have metastatic castration-resistant prostate cancer (mCRPC) that progressed after novel antiandrogen therapy and have not received chemotherapy for mCRPC. **Secondary Objectives:** To evaluate Prostate Specific Antigen Response Rate (PSA-RR), Objective Response Rate, and Time to Response.

2.2 Background & Rationale To address the need for improved therapeutic outcomes in patients with mCRPC that has progressed after a second-generation hormonal manipulation (novel antiandrogen therapy [NAT]). Standard-of-care chemotherapy (docetaxel) has limited long-term durability. This study investigates whether the addition of a next-generation anti-CTLA-4 agent (BMS-986218), with or without a PD-1 inhibitor (nivolumab), can synergize with docetaxel to overcome an immunosuppressive tumor microenvironment and improve survival outcomes compared to docetaxel alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Docetaxel alone) **Blinding:** Open-label **Randomization:** Randomized **Overall Status:** Terminated

3.2 Planned Enrollment & Duration Targeted Enrollment: 10 participants **Number of Sites:** ~31 global locations **Estimated Study Start:** 14-Feb-2022 **Estimated Primary Completion:** Dec-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants with histologic confirmation of carcinoma of the prostate without small cell features. 2. Documented prostate cancer progression by Prostate Cancer Working Group 3 (PCWG3) criteria while castrate. 3. Evidence of metastatic disease documented by either bone lesions on radionuclide bone scan and/or soft tissue lesions on computed tomography (CT)/magnetic resonance imaging (MRI). 4. Eastern Cooperative Oncology Group (ECOG) performance status 0 to 1. 5. Ongoing androgen deprivation therapy (ADT) with a gonadotropin-releasing hormone (GnRH) agonist/antagonist or bilateral orchiectomy (i.e., surgical or medical castration) confirmed by testosterone level ≤ 1.73 nmol/L (50 ng/dL) at the screening visit. 6. Chemotherapy-naïve for metastatic castration-resistant prostate cancer (mCRPC) and have received at least one novel antiandrogen therapy (NAT).

4.2 Exclusion Criteria 1. Concurrent malignancy (present during screening) requiring treatment or history of prior malignancy active within 2 years prior to treatment assignment in Part 1 or randomization in Part 2. 2. Untreated central nervous system (CNS) metastases. 3. Leptomeningeal metastases. 4. Active, known, or suspected autoimmune disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Docetaxel (standard dosing) administered alone, or combined with intravenous BMS-986218, or combined with both BMS-986218 and Nivolumab in 3-week cycles. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Up to 2 years of treatment, or until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Ongoing androgen deprivation therapy (ADT) with a GnRH agonist/antagonist. Daily prednisone alongside Docetaxel. **Prohibited:** Concurrent investigational agents, other systemic anticancer therapies, and live vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Imaging (CT/MRI/Bone Scan), Testosterone confirmation
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 3 Weeks	Treatment administration, Safety Labs, AE Assessment
Imaging Follow-up	Weeks 9, 18, 27	Tumor assessments every 9 weeks ($\pm$ 7 days) for the first 27 weeks, then every 12 weeks
End of		

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Number of Participants With Treatment Related Adverse Events:** Evaluated using National Cancer Institute Common Terminology Criteria for Adverse Events version 5 (NCI CTCAE v5).
- 2. Number of Participants With Treatment Related Serious Adverse Events:** Evaluated using NCI CTCAE v5.
- 3. Number of Participants With Dose Limiting Toxicities (DLTs):** Defined as:
 - Any treatment-related AEs for which a participant permanently discontinues a study treatment (other than daily prednisone) and that occurs during the first 2 cycles of treatment.
 - Any death not clearly due to the underlying disease or extraneous causes and that occurs during the first 2 cycles of treatment.
 - Greater than or equal to Grade 2 pneumonitis lasting greater than 5 days despite appropriate medical therapy and that occurs during the first 2 cycles of treatment.
 - Any neutropenic fever as well as Grade 4 neutropenia or thrombocytopenia for > 7 days that occurs during the first 2 cycles of treatment.
 - Any treatment-related AE that delays initiation of Cycle 2 or Cycle 3 of treatment by greater than 2 consecutive weeks.

7.2 Secondary Outcomes

1. Prostate Specific Antigen Response Rate (PSA-RR)
2. Objective Response Rate
3. Time to Response

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Graded and reported using the National Cancer Institute Common Terminology Criteria for Adverse Events version 5 (NCI CTCAE v5). **Serious Adverse Events (SAEs):** Reported promptly according to standard regulatory timelines. DLTs evaluate severe events like Grade ≥ 2 pneumonitis or Grade 4 neutropenia/thrombocytopenia lasting > 7 days. **Data Monitoring Committee (DMC):** Utilized a Bayesian Optimal Interval (BOIN) design for the safety lead-in evaluation to monitor toxicity thresholds.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All randomized participants. **Sample Size Justification:** Utilized adaptive BOIN design for dose-escalation/safety lead-in (enrollment in cohorts of 3 to 4 participants). Overall enrollment analyzed was 10 patients. **Handling of Missing Data:** Standard time-to-event methodologies (e.g., Kaplan-Meier estimates) utilized for trial endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine remote and onsite monitoring visits to evaluate investigator compliance and protocol adherence.

Audit Trail: Robust clinical data management including eCRF locking, data clarification, and Blinded Independent Central Review (BICR) for imaging endpoints.

11. Study Identifiers & Governance

Primary ID: CA022-009 **Posting ID:** 987628428492287200 **Registry**

Number: NCT05169684 **Sponsor/Lead Organization:** Bristol Myers

Squibb Study Title: A Study of BMS-986218 or BMS-986218 Plus Nivolumab in Combination With Docetaxel in Participants With

Metastatic Castration-resistant Prostate Cancer **Official**

Regulatory Title: A Phase 2, Open-label, Randomized Controlled Trial of BMS-986218 or BMS-986218 Plus Nivolumab in Combination With Docetaxel in Participants With Metastatic Castration-

resistant Prostate Cancer **Trial Status:** TERMINATED (Study was terminated early)

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Preferred UMLS Name:** Prostatic Neoplasms,

Castration-Resistant **Diagnosis Threshold:** Disease progression by PCWG3 criteria with castrate testosterone levels ≤ 50 ng/dL

Medical Methodology: Immune checkpoint blockade (anti-CTLA-4 + anti-PD-1) combined with standard-of-care cytotoxic

chemotherapy **Optimization Target:** Disease Control and Safety

13. Study Procedures & Methodology

Management Pathway: Docetaxel-based chemotherapy integrated with next-generation immunotherapy infusions **Education Protocol (HCP):**

Protocol-specific safety management for immune-related adverse events (irAEs) **Education Protocol (Patient):** Informed consent

detailing compounding toxicities of chemotherapy and

immunotherapy **Quality Audit Mechanism:** Central imaging review (BICR) and source data verification

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years of active treatment, followed by long-term survival follow-up **Onsite Visit Cadence:** Every 3 weeks

(Q3W) for treatment cycles **Remote Monitoring Frequency:** Routine

survival tracking post-treatment discontinuation **Checklist Review**
Interval: Scans evaluated every 9 weeks for the first 27 weeks,
then every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date					
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					